

Aircraft manufacturing: Stratolaunch Systems, Mojave California, USA — New Megadoor installation

ASSA ABLOY Entrance Systems provides Stratolaunch with a Megadoor for the worlds largest aircraft.

ASSA ABLOY Entrance Systems proudly provided Stratolaunch Systems, a Paul G. Allen Project, with a 420'w x 68'h (128m x 21m) hangar door system for their new fabrication facility located in Mojave California. Inside this facility, the world's largest aircraft is being fabricated by Scaled Composites which has a wingspan of 380' (116m) and thrust provided by (6) 747 aircraft engines. This aircraft will be used as a carrier vehicle, flying to 30,000ft with a rocket that will then be launched with a payload destined for space. This system will revolutionize space transportation.

In 2004, SpaceShipOne ushered in a new era of space travel, when it became the first non-governmental manned rocket ship to fly beyond the earth's atmosphere. SpaceShipOne was developed by Mojave Aerospace, a joint venture of investor and philanthropist Paul Allen, and Scaled Composites, a Mojave-based aerospace company founded by legendary aerospace engineer Burt Rutan. SpaceShipOne became the first private manned spacecraft to exceed 328,000 feet twice in 14 days, earning it the prestigious $10 million Ansari X-Prize. Business entrepreneur Richard Branson of Virgin Group has since licensed the technology behind SpaceShipOne for Virgin Galactic, a venture that will take paying customers into suborbital space.

Hangar Statistics:
- 103 257sq ft. (9 592 m²)
- 420' clear span (128m)
- 3,000,000lbs of structural steel
- 97' ridge line (29,5m)

End user:	Stratolaunch Systems www.stratolaunch.com
Design build contractor:	Wallace & Smith General Contractors www.wallacesmith.com
Metal building supplier:	CBC Steel Buildings www.cbcsteelbuildings.com

Critical issues:
Desert Conditions: The composite carrier vehicle being crafted inside by the highly skilled technicians requires an environment protected from the harsh conditions of the Mojave desert. For example, the fine dust blowing around the desert airport is notorious for coating everything not properly protected. Keeping sensitive aviation electronics, engines and other aircraft components from being affected by the dust and sand can be a costly challenge. As proven in previous Mojave Airport installations, the unique perimeter seals and low air infiltration values of the **Megadoor** are more effective than a traditional sliding doors design in sealing out this fine desert dust. Furthermore, the doors do not require a bottom rail system which requires regular cleaning.

Climate Control: In order to increase worker comfort and productivity in the hot desert summers, a combination of exhaust fans and evaporator coolers are utilized. The client wanted the doors be closed during normal operation for privacy, access and environmental control. In order to meet this requirement, the project Mechanical Engineer specified four massive exhaust fans be incorporated into the **Megadoor system**. These provide ground level air movement for the cross flow ventilation strategy complementing the fans on the top of the hangar on the opposite wall.

Natural Lighting: Translucent fabric was specified on the **Megadoor systems** to allow ten percent of the natural light into the hangar. The soft, diffused light creates a great work environment for the technicians and reduces lighting costs.

ASSA ABLOY Entrance Systems provided:
- **Megadoor hangar door** design and assistance the structural integration of the door system into the building
- One 420'w × 68'h (128m x 21m) **Megadoor** system consisting of 7 individually operated door leafs and 6 retractable mullions
- Upgraded Control System: Including touchscreen user interface, windows based browser, multilevel password protection, event and alarm log and service monitoring tools.
- Installation Supervision



ASSA ABLOY Entrance Systems is a leading supplier of entrance automation solutions for efficient flow of goods and people. With our worldwide presence we offer products and services dedicated to satisfying end-user needs for safe, secure, convenient and sustainable operations. ASSA ABLOY Entrance Systems is a division within ASSA ABLOY.

ASSA ABLOY

www.assaabloyentrance.com